Ehsan Sharafian

Ph.D. Candidate, Department of Mechanical Engineering, University of Maine, U.S.

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PROFESSIONAL SUMMARY

PhD candidate in engineering with 6+ years of research and industry experience, seeking a Summer 2027 internship in wearable technology. Experienced in developing wearable systems for human movement analysis and rehabilitation, integrating IMU sensors, machine learning, embedded hardware, smartphone applications, and haptic feedback. Author of six peer-reviewed publications.

TECHNICAL SKILLS

Programming & ML: Python, MATLAB, C/C++, Java, NumPy, Scikit-learn, PyTorch, CNNs

Engineering Tools: SolidWorks, Simulink, PLC programming, Android Studio, Git

Wearable Systems: IMUs, sensor fusion, embedded systems, Arduino, PCB modules, SparkFun modules

Motion Analysis: OpenCV, OpenSim, gait analysis, activity recognition, feature extraction, SPSS

RESEARCH EXPERIENCE

Graduate Research Assistant | **Biorobotics and Biomechanics Lab, University of Maine, U.S.** **Sep 2023 - Present**

- Developed a gait monitoring system using IMU, implemented on a smartphone, achieving 96% walking condition classification accuracy and <1.1 cm foot trajectory error in real time.
- Developed an activity recognition classifier for older adults, achieving 99% accuracy for normal locomotion and 93% for atypical stair patterns.
- Built a haptic gait training prototype with embedded hardware and an Android application, improving walking speed by up to 30% with minimal cognitive demand in older adults.

Research Assistant | **Amirkabir University of Technology** **Sep 2018 - Sep 2020**

- Developed spot-welding trajectory optimization methods using GA and PSO, achieving 24% shorter computed cycle times for automotive production.
- Developed a screw theory formulation to evaluate joint load distribution in a Delta robot for efficient force transmission.

INDUSTRY EXPERIENCE

Robotics Engineer | **Sina Robotics and Medical Innovators Co.** **Dec 2022 - Sep 2023**

- Developed PLC control logic and validated inverse kinematics in Simulink for a telesurgical laparoscopic robotic platform.
- Built a Python GUI on Raspberry Pi, incorporating feedback from 6 surgeons through direct communication during testing.

Junior R&D Engineer | **Saipa Automotive Manufacturing Company** **Jul 2018 - Jul 2019**

- Modeled robot kinematics and dynamics to assess trajectory feasibility for an automotive welding workstation.
- Collaborated with R&D to evaluate robot trajectories against workstation constraints and identify efficiency improvements.

ENGINEERING PROJECTS

- Integrated OpenCV to track foot motion and analyzed gait mat data to study obstacle crossing mechanisms. **Jun 2026**
- Developed a CNN for gait assessment across 12 walking speeds, achieving 95% validation accuracy. **Dec 2023**
- Modeled serial robot dynamics using neural networks and optimized trajectories using heuristic algorithms. **Dec 2022**
- Implemented adaptive and disturbance observer based control for attitude tracking of parallel manipulators. **May 2021**

TEACHING EXPERIENCE

Teaching Assistant, Manufacturing Design | **University of Maine, U.S.** **Sep 2023 - Dec 2023**

- Instructed students in SolidWorks for Manufacturing Design, guiding CAD modeling and mechanisms design projects.

EDUCATION

Ph.D. in Mechanical Engineering | **University of Maine, U.S.** | GPA: 4.00/4.00 **Sep 2023 - Exp Dec 2027**

M.S. in Mechanical Engineering | **Amirkabir University of Technology** | GPA: 3.65/4.00 **Sep 2017 - Feb 2020**

B.S. in Mechanical Engineering | **University of Tehran** | GPA: 3.15/4.00 **Sep 2012 - Jul 2017**

PUBLICATIONS

Author of 6 peer-reviewed papers (5 first author) in digital health, rehabilitation, and robotics, listed on Google Scholar.